1 **(a)** Define the radian.

.....

..... [1]

(b) A circular metal disc spins horizontally about a vertical axis, as shown in Fig. 1.1.

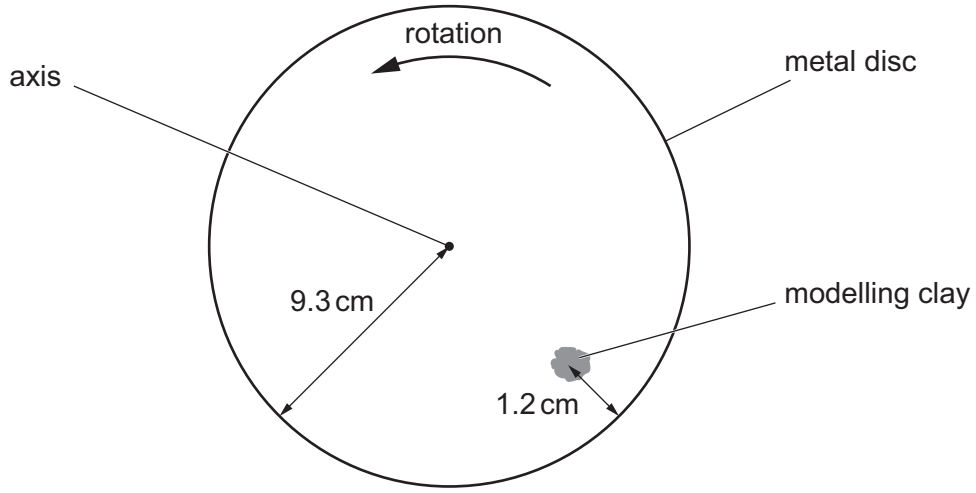


Fig. 1.1 (not to scale)

A piece of modelling clay is attached to the disc.

For the instant when the piece of modelling clay is in the position shown, draw on Fig. 1.1:

(i) an arrow, labelled V, showing the direction of the velocity of the modelling clay [1]

(ii) an arrow, labelled A, showing the direction of the acceleration of the modelling clay. [1]

(c) The metal disc in Fig. 1.1 has a radius of 9.3 cm.
The centre of gravity of the modelling clay is 1.2 cm from the rim of the disc and moves with a speed of 0.68 m s^{-1}.

(i) Calculate the angular speed ω of the disc.

ω = ... rad s^{-1} [2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(ii) Calculate the acceleration a of the centre of gravity of the modelling clay.

a = ... $m\,s^{-2}$ [2]

(d) A second piece of modelling clay is attached to the disc in the position shown in Fig. 1.2.

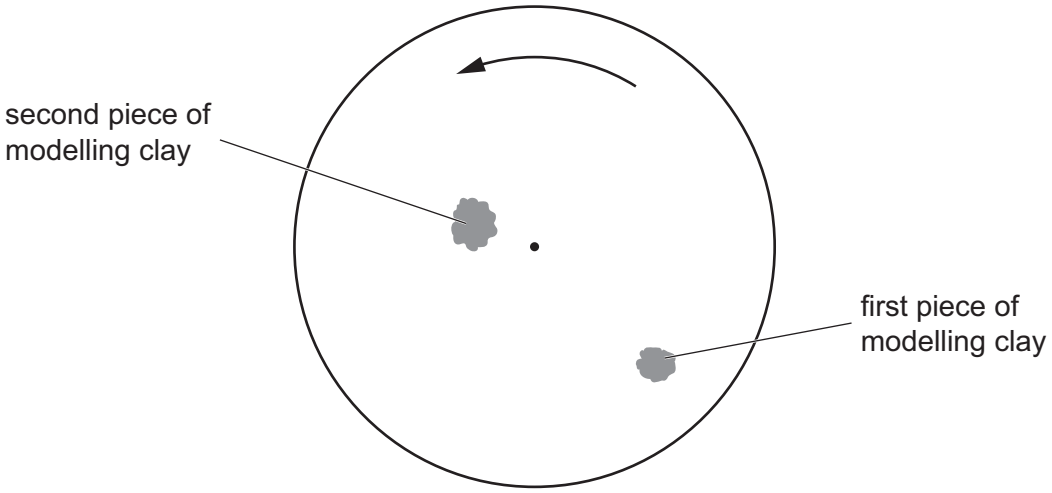


Fig. 1.2

The second piece of modelling clay has a larger mass than the first piece.

By placing **one** tick (✓) in each row, complete Table 1.1 to show how the quantities indicated compare for the two pieces of modelling clay.

Table 1.1

quantity	less for second piece than first piece	same for both pieces	greater for second piece than first piece
angular speed			
linear speed			
acceleration			

[3]

[Total: 10]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

Markscheme — 9702/42/M/J/24 · Q1

Disc with modelling clay — angular speed & centripetal acceleration

(a) [1 mark]

The angle subtended at the centre of a circle by an arc whose length is **equal to the radius**.

(b)(i) [1 mark]

Draw an arrow labelled V from the modelling clay pointing in the **NE direction** — the velocity is tangential to the circular path (perpendicular to the radius of the disc).

(b)(ii) [1 mark]

Draw an arrow labelled A from the modelling clay pointing in the **NW direction**, i.e. towards the centre of the disc (the direction of the centripetal acceleration).

(c)(i) [2 marks]

Distance of the clay from the axis: $r = 9.3 - 1.2 = 8.1$ cm $= 8.1\times10^{-2}$ m
$v = r\omega$
$\omega = v/r = 0.68 / 8.1\times10^{-2} =$ **8.4 rad s^{-1}**

(c)(ii) [2 marks]

$a = v^2/r = 0.68^2 / 8.1\times10^{-2} = 0.4624 / 8.1\times10^{-2}$
$=$ **5.7 m s^{-2}**

(d) [3 marks]

angular speed: **same for both pieces** — the disc is one rigid body, so every point has the same ω.
linear speed: **less for the second piece** — $v = r\omega$ and the second piece is nearer the axis.
acceleration: **less for the second piece** — $a = r\omega^2$ and the second piece has the smaller radius.

1 A steel ball is placed on the inside surface of a hollow circular cone. The ball moves in a horizontal circle at constant speed, as shown in Fig. 1.1.

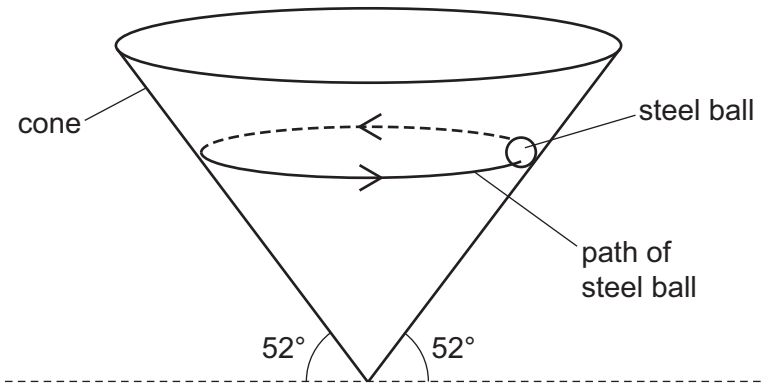


Fig. 1.1

The angle of the side of the cone to the horizontal is 52°. There is no friction between the ball and the cone.

(a) Fig. 1.2 shows a cross-section through the cone and the steel ball.

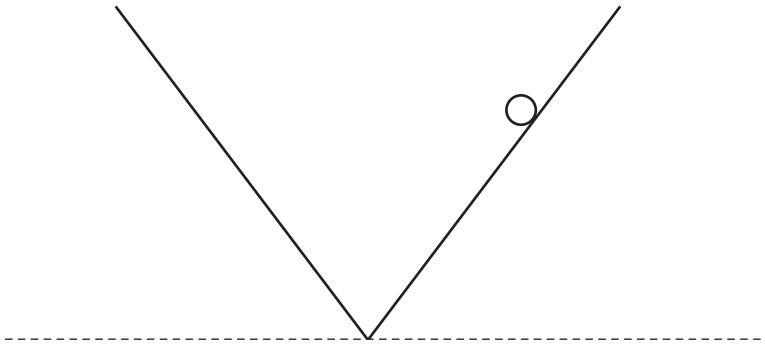


Fig. 1.2

On Fig. 1.2, draw labelled arrows to show the **two** forces acting on the ball. [1]

(b) Describe how the forces acting on the ball cause its acceleration to be centripetal.

.....

.....

.....

..... [2]

© UCLES 2025

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(c) The ball moves in a circle of radius 0.15 m.

Show that the speed of the ball is 1.4 m s^{-1}.

[3]

(d) Calculate the angular speed ω of the ball.

ω = ... rad s^{-1} [2]

(e) The speed of the ball is increased.

Explain why the radius of the circular path of the ball increases.

.....

.....

..... [1]

[Total: 9]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

Markscheme — 9702/42/F/M/25 · Q1

Steel ball inside a hollow cone — circular motion & angular speed

(a) [1 mark]

An arrow **vertically downwards** labelled ‘weight’, and an arrow **perpendicular to the cone surface, pointing inwards and upwards**, labelled ‘normal contact force’ (the two arrows must be in the correct directions and correctly labelled).

(b) [2 marks]

The **vertical component of the contact force equals the weight**, so there is no resultant force vertically. The **horizontal component of the contact force is the resultant force**, and it acts towards the centre of the circle — this provides the centripetal force that keeps the ball moving in a circle.

(c) [3 marks]

For a conical-pendulum type of motion the acceleration is **a = g tan 52°**, and $a = v^2/r$:
$9.81 \times \tan 52° = v^2/0.15$
$v^2 = 0.15 \times 9.81 \times \tan 52° = 1.88\ m^2\ s^{-2}$
$v =$ **$1.4\ m\ s^{-1}$**

(d) [2 marks]

$v = r\omega$, so $\omega = v/r = 1.4/0.15$
$=$ **$9.3\ rad\ s^{-1}$**
(Equivalently $\omega = \sqrt{(g \tan 52°/0.15)} = 9.3\ rad\ s^{-1}$.)

(e) [1 mark]

The resultant (centripetal) force — hence the acceleration — is unchanged, and $a = v^2/r$, so **v^2 is proportional to r**. Therefore increasing the speed requires a larger radius, so the ball moves up the cone to a circle of larger radius.

1 **(a)** Define the radian.

.....

..... [1]

(b) The rear wheel and the pedals of a bicycle are connected by a chain that passes around two cogs (toothed wheels), as shown in Fig. 1.1.

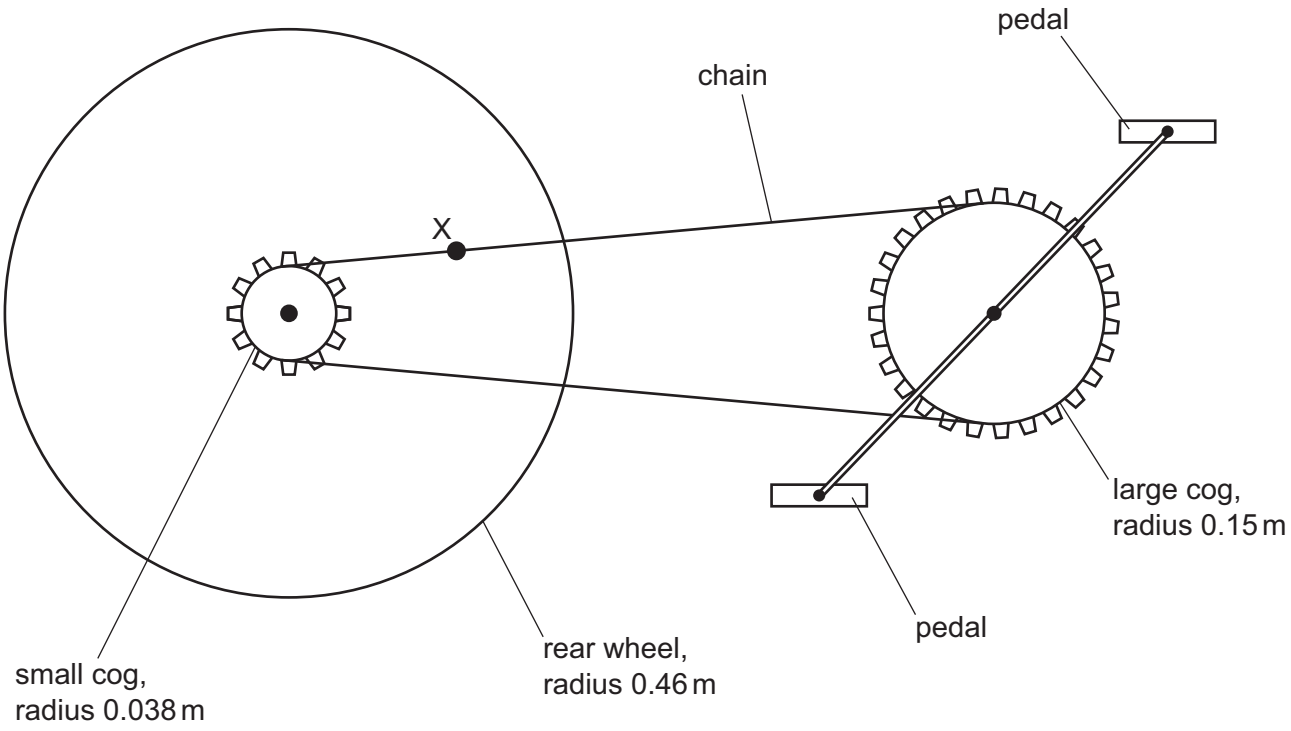


Fig. 1.1 (not to scale)

The small cog has a radius of 0.038 m and is fixed to the rear wheel so that it rotates with it. The large cog has a radius of 0.15 m and is fixed to the pedals so that it rotates with them. The rear wheel has a radius of 0.46 m.

The bicycle is being pedalled so that it moves in a straight line at a constant speed of 17 m s^{-1}.

(i) Calculate the angular speed of the rear wheel.

angular speed = ... rad s^{-1} [2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(ii) Calculate the period of rotation of the small cog.

period = ... s [2]

(iii) Show that the distance moved by point X on the chain during one full rotation of the small cog is 0.24 m.

[1]

(iv) Use the information in **(b)(iii)** to determine the angle through which the large cog rotates during one full rotation of the small cog.

angle = ... rad [2]

(c) The chain of the bicycle in **(b)** is moved onto a smaller cog fixed to the rear wheel. The speed of the bicycle does not change.

Explain, without calculation, the effect of this change on the angular speed of the pedals.

.....

.....

..... [2]

[Total: 10]

DO NOT WRITE IN THIS MARGIN

Markscheme — 9702/42/M/J/25 · Q1

Bicycle chain & cogs — radian, angular speed & period

(a) [1 mark]

The **angle subtended at the centre of a circle by an arc whose length is equal to the radius**.

(b)(i) [2 marks]

$v = r\omega$, so $\omega = v/r = 17/0.46$
$= \mathbf{37\ rad\ s^{-1}}$

(b)(ii) [2 marks]

$T = 2\pi r/v = 2\pi \times 0.46/17$ (or $T = 2\pi/\omega = 2\pi/37$)
$= \mathbf{0.17\ s}$

(b)(iii) [1 mark]

distance moved = circumference of the small cog $= 2\pi r = 2\pi \times 0.038$
$= \mathbf{0.24\ m}$

(b)(iv) [2 marks]

angle = arc length/radius = 0.24/0.15
$= \mathbf{1.6\ rad}$

(c) [2 marks]

The rear wheel still rotates at the same angular speed (the bicycle speed is unchanged), but the rear cog is now smaller, so **point X on the chain moves through a smaller distance in the same time** — the linear speed of the chain decreases. The large cog (fixed to the pedals) has the same radius, so the **angular speed of the pedals decreases**.

1 **(a)** In terms of velocity and acceleration, describe uniform circular motion of an object.

.....

.....

..... [2]

(b) Fig. 1.1 shows the view from above of a polystyrene ball undergoing horizontal circular motion of radius *R*.

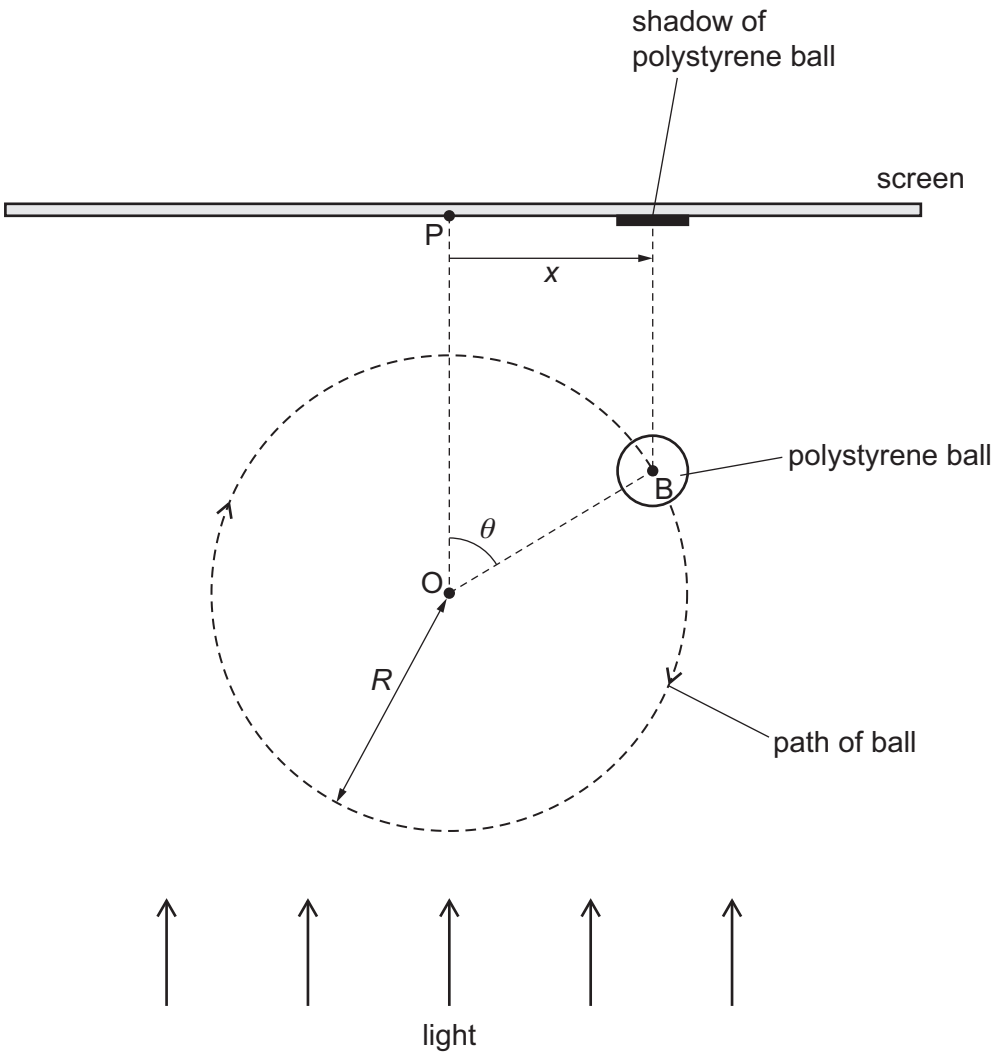


Fig. 1.1

The ball is illuminated by parallel light so that a shadow of the ball forms on a screen placed on the opposite side of the ball from the light source.

The line joining points O and P is perpendicular to the screen.

The angular speed of the circular motion is ω.

DO NOT WRITE IN THIS MARGIN

(i) State an expression, in terms of R and ω, for the speed v of the ball.

v = ... [1]

(ii) Determine an expression, in terms of v and ω, for the centripetal acceleration of the ball.

centripetal acceleration = ... [2]

(c) The ball in **(b)** is in the position shown in Fig. 1.1, such that line OB is at an angle θ to the line OP.

(i) Determine an expression, in terms of R and θ, for the displacement x of the shadow from P.

x = ... [1]

(ii) The value of θ is zero at time $t = 0$.

State an expression for θ in terms of ω and t.

θ = ... [1]

(iii) Use your answers in **(c)(i)** and **(c)(ii)** to show that x is given by

$$x = R\sin\omega t.$$

[1]

(iv) Explain, with reference to the equation in **(c)(iii)**, why the motion of the shadow of the ball on the screen may be modelled as simple harmonic.

.....

..... [1]

DO NOT WRITE IN THIS MARGIN

(d) The circular motion of the ball in Fig. 1.1 has a diameter of 0.46 m and an angular speed of 1.9 rad s^{-1}.

For the simple harmonic motion of the shadow of the ball in Fig. 1.1, calculate:

(i) the amplitude

amplitude = ... m [1]

(ii) the period

period = ... s [2]

(iii) the maximum acceleration.

maximum acceleration = ... $m\,s^{-2}$ [2]

(e) On Fig. 1.1, draw, and label with the letter A, the position of the shadow on the screen when the shadow has its maximum positive acceleration. [1]

[Total: 15]

DO NOT WRITE IN THIS MARGIN

Markscheme — 9702/41/O/N/25 · Q1

Circular motion & simple harmonic motion — shadow projection

(a) [2 marks]

The velocity and acceleration both have **constant magnitude**, and the velocity is **always perpendicular to the acceleration** (the velocity is tangential and the acceleration is directed towards the centre).

(b)(i) [1 mark]

$v = \mathbf{R\omega}$

(b)(ii) [2 marks]

$a = R\omega^2 = (R\omega)\omega = \mathbf{v\omega}$

(c)(i) [1 mark]

$x = \mathbf{R \sin \theta}$

(c)(ii) [1 mark]

$\theta = \mathbf{\omega t}$

(c)(iii) [1 mark]

Substituting $\theta = \omega t$ into $x = R \sin \theta$ gives $\mathbf{x = R \sin \omega t}$.

(c)(iv) [1 mark]

The equation is of the form $\mathbf{x = x_0 \sin \omega t}$, the defining equation of simple harmonic motion (with amplitude $x_0 = R$), so the shadow's motion is simple harmonic.

(d)(i) [1 mark]

amplitude = diameter/2 = 0.46/2 = **0.23 m**

(d)(ii) [2 marks]

$\omega = 2\pi/T$, so $T = 2\pi/\omega = 2\pi/1.9$
$= \mathbf{3.3\ s}$

(d)(iii) [2 marks]

$a_0 = \omega^2 x_0 = 1.9^2 \times 0.23$
$= \mathbf{0.83\ m\ s^{-2}}$

(e) [1 mark]

The shadow is at the extreme of its motion — at the **left-hand edge of the circular path** (maximum displacement from P), labelled A, where the acceleration is a maximum.

1 The Earth may be considered as a uniform sphere of radius 6.37×10^6 m.

Cambridge is at a point on the Earth's surface that has a latitude of 52.2° north of the Equator, as shown in Fig. 1.1.

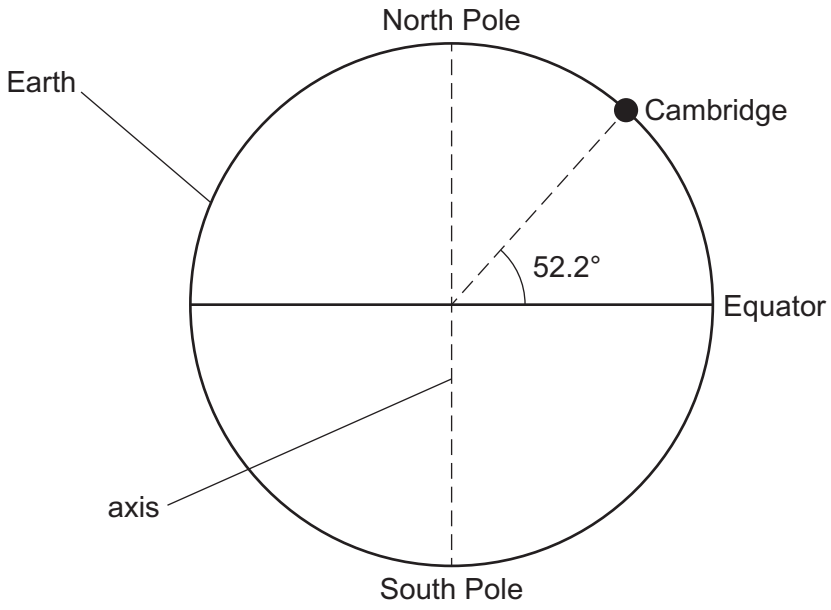


Fig. 1.1

As the Earth spins on its axis, Cambridge moves in a circle that is parallel to the Equator but with a smaller radius.

(a) (i) Show that the radius of the circle around which Cambridge moves is 3.90×10^6 m.

[1]

(ii) Calculate the speed at which Cambridge moves around the circle.

speed = ... $m\,s^{-1}$ [3]

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(b) A student of mass 58.6 kg stands on horizontal ground in Cambridge.

(i) Determine the magnitude of the resultant force that acts to cause the circular motion of the student.

resultant force = ... N [2]

(ii) On Fig. 1.2, draw an arrow to show the direction of the resultant force that acts on the student.

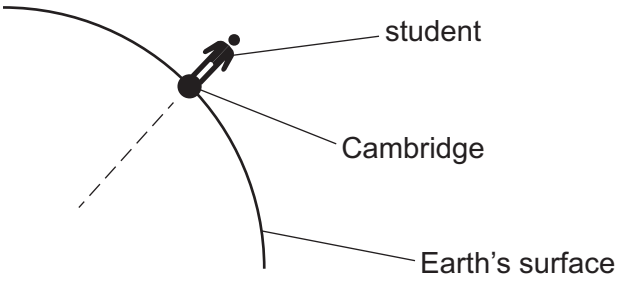


Fig. 1.2 (not to scale)

[1]

(iii) On Fig. 1.3, draw labelled arrows from the student to show the directions of the forces that act on the student to cause the resultant force in **(b)(ii)**.

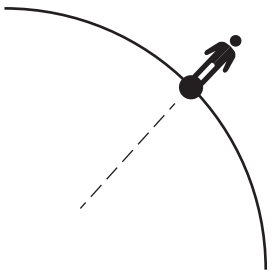


Fig. 1.3 (not to scale)

[2]

[Total: 9]

Markscheme — 9702/42/O/N/25 · Q1

Circular motion of a student at latitude 52.2°

(a)(i) [1 mark]

radius = $6.37 \times 10^6 \times \cos 52.2°$
= **3.90×10^6 m**

(a)(ii) [3 marks]

$v = 2\pi r/T = 2\pi \times 3.90 \times 10^6/(24 \times 60 \times 60)$
= **280 m s^{-1}**

(b)(i) [2 marks]

$F = mv^2/r = 58.6 \times 280^2/(3.90 \times 10^6)$
= **1.2 N**

(b)(ii) [1 mark]

An arrow pointing **horizontally to the left** (towards the Earth's axis of rotation).

(b)(iii) [2 marks]

An arrow **downwards** from the student (towards the centre of the Earth) labelled 'weight', and an **upwards** arrow (the normal contact force) pointing slightly **to the left of the normal**, labelled 'contact force' — their resultant is the horizontal centripetal force.

1 **(a)** In terms of velocity and acceleration, describe uniform circular motion of an object.

.....

.....

..... [2]

(b) Fig. 1.1 shows the view from above of a polystyrene ball undergoing horizontal circular motion of radius *R*.

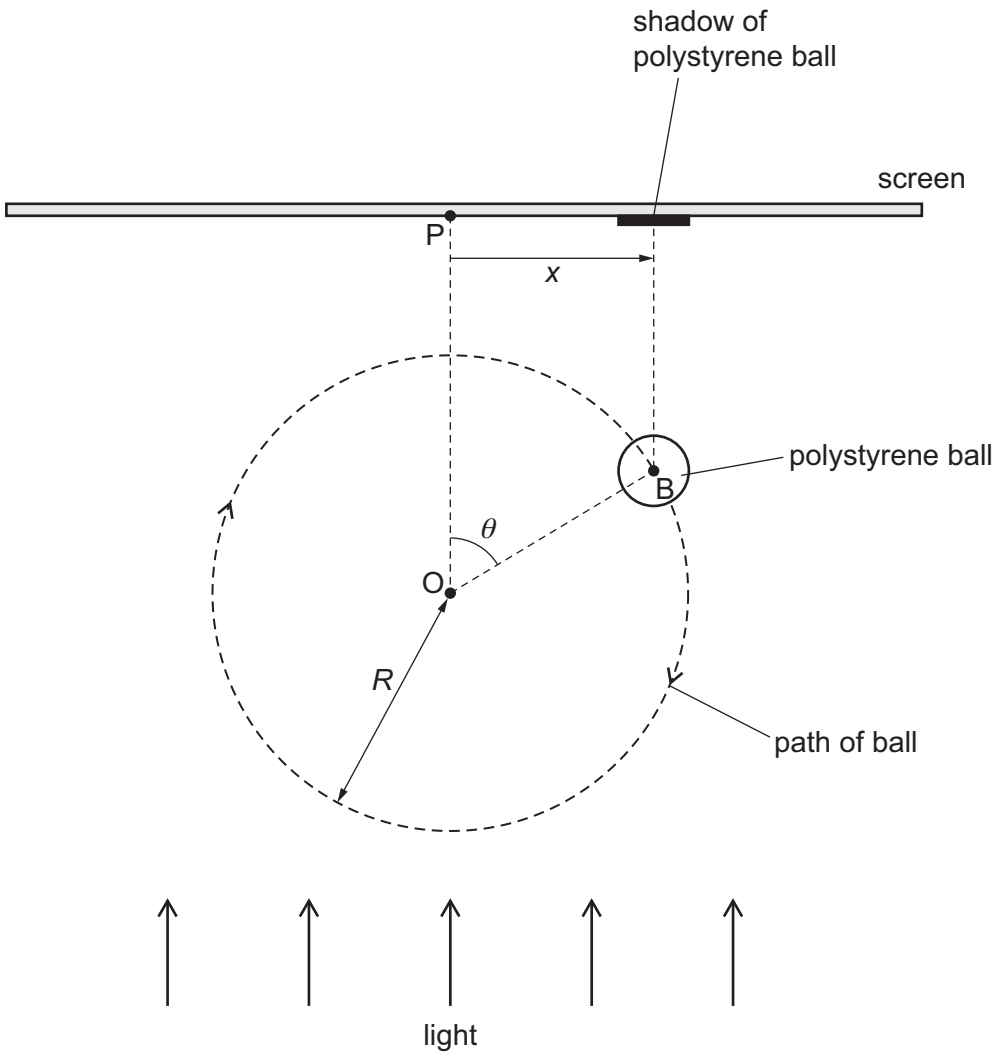


Fig. 1.1

The ball is illuminated by parallel light so that a shadow of the ball forms on a screen placed on the opposite side of the ball from the light source.

The line joining points O and P is perpendicular to the screen.

The angular speed of the circular motion is ω.

DO NOT WRITE IN THIS MARGIN

(i) State an expression, in terms of R and ω, for the speed v of the ball.

v = ... [1]

(ii) Determine an expression, in terms of v and ω, for the centripetal acceleration of the ball.

centripetal acceleration = ... [2]

(c) The ball in **(b)** is in the position shown in Fig. 1.1, such that line OB is at an angle θ to the line OP.

(i) Determine an expression, in terms of R and θ, for the displacement x of the shadow from P.

x = ... [1]

(ii) The value of θ is zero at time $t = 0$.

State an expression for θ in terms of ω and t.

θ = ... [1]

(iii) Use your answers in **(c)(i)** and **(c)(ii)** to show that x is given by

$$x = R\sin\omega t.$$

[1]

(iv) Explain, with reference to the equation in **(c)(iii)**, why the motion of the shadow of the ball on the screen may be modelled as simple harmonic.

.....

..... [1]

DO NOT WRITE IN THIS MARGIN

(d) The circular motion of the ball in Fig. 1.1 has a diameter of 0.46 m and an angular speed of 1.9 rad s^{-1}.

For the simple harmonic motion of the shadow of the ball in Fig. 1.1, calculate:

(i) the amplitude

amplitude = ... m [1]

(ii) the period

period = ... s [2]

(iii) the maximum acceleration.

maximum acceleration = ... $m\,s^{-2}$ [2]

(e) On Fig. 1.1, draw, and label with the letter A, the position of the shadow on the screen when the shadow has its maximum positive acceleration. [1]

[Total: 15]

DO NOT WRITE IN THIS MARGIN

Markscheme — 9702/43/O/N/25 · Q1

Circular motion & simple harmonic motion — shadow projection

(a) [2 marks]

The velocity and acceleration both have **constant magnitude**, and the velocity is **always perpendicular to the acceleration** (the velocity is tangential and the acceleration is directed towards the centre).

(b)(i) [1 mark]

$v = \mathbf{R\omega}$

(b)(ii) [2 marks]

$a = R\omega^2 = (R\omega)\omega = \mathbf{v\omega}$

(c)(i) [1 mark]

$x = \mathbf{R \sin \theta}$

(c)(ii) [1 mark]

$\theta = \mathbf{\omega t}$

(c)(iii) [1 mark]

Substituting $\theta = \omega t$ into $x = R \sin \theta$ gives $\mathbf{x = R \sin \omega t}$.

(c)(iv) [1 mark]

The equation is of the form $\mathbf{x = x_0 \sin \omega t}$, the defining equation of simple harmonic motion (with amplitude $x_0 = R$), so the shadow's motion is simple harmonic.

(d)(i) [1 mark]

amplitude = diameter/2 = 0.46/2 = **0.23 m**

(d)(ii) [2 marks]

$\omega = 2\pi/T$, so $T = 2\pi/\omega = 2\pi/1.9$
$= \mathbf{3.3\ s}$

(d)(iii) [2 marks]

$a_0 = \omega^2 x_0 = 1.9^2 \times 0.23$
$= \mathbf{0.83\ m\ s^{-2}}$

(e) [1 mark]

The shadow is at the extreme of its motion — at the **left-hand edge of the circular path** (maximum displacement from P), labelled A, where the acceleration is a maximum.

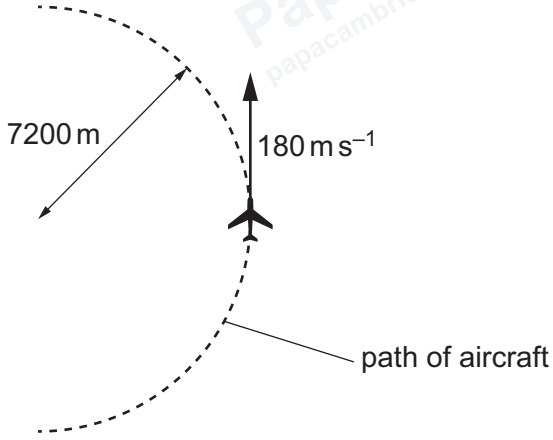
1 **(a)** With reference to velocity and acceleration, describe uniform circular motion.

.....

.....

..... [2]

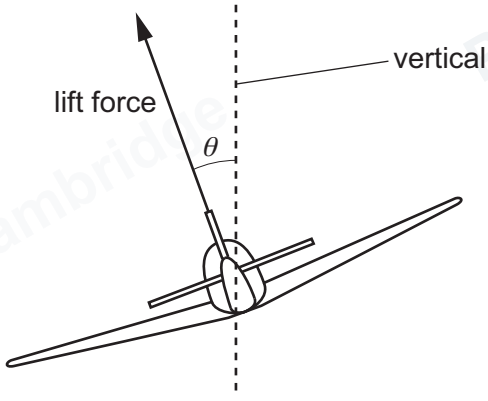
(b) An aircraft is travelling in a horizontal circle of radius 7200 m at a speed of 180 m s^{-1}, as shown in the view from above in Figure 1.1.



not to scale

Figure 1.1

In order to move in a circle, the aircraft has to tilt its wings so that the lift force from the wings is not vertical, as shown in the view from behind in Figure 1.2.



not to scale

Figure 1.2

(i) With reference to the forces acting on the aircraft, explain why the circular motion of the aircraft requires the lift force to be in the direction shown rather than vertically upwards.

.....

.....

..... [2]

(ii) On Figure 1.2, draw an arrow from the aircraft to represent the direction of the resultant force on the aircraft. [1]

(iii) Calculate the centripetal acceleration of the aircraft.

acceleration = ... $m\,s^{-2}$ [2]

(iv) Determine the angle θ at which the aircraft is tilted. Explain your reasoning.

θ = ... ° [3]

[Total: 10]

PapaCambridge
papacambridge.com

Markscheme — 9702/42/M/J/26 · Q1

Aircraft in horizontal circle — centripetal acceleration & banking angle

(a) [2 marks]

The acceleration is (always) perpendicular to the velocity, directed towards the centre of the circle, and the magnitudes of the velocity (the speed) and of the acceleration are constant — but their **directions change continuously**.

(b)(i) [2 marks]

The only other force on the aircraft is its weight, which is vertical and has no horizontal component. The lift must therefore have a horizontal component (the resultant force towards the centre of the circle) to provide the centripetal force — so the lift **cannot be vertically upwards**.

(b)(ii) [1 mark]

Arrow drawn from the aircraft pointing **horizontally to the left**, i.e. towards the centre of the circle — the resultant force provides the centripetal force.

(b)(iii) [2 marks]

$a = v^2/r = 180^2/7200$
$= \mathbf{4.5\ m\ s^{-2}}$

(b)(iv) [3 marks]

The vertical component of the lift balances the weight ($L \cos \theta = mg$) and the horizontal component of the lift causes the centripetal acceleration ($L \sin \theta = ma$):
$\tan \theta = a/g = 4.5/9.81$
$\theta = \mathbf{25°}$